

## Multiple Choice

1. **Answer:** B

*Options:* A) The resonance pulls Io in different directions and generates heat.; B) It makes the orbit of Io slightly elliptical.; C) It creates a gap with no asteroids between the orbits.; D) It prevents formation of the ring material into other moons.

*Note:* Correct option: B - It makes the orbit of Io slightly elliptical.

2. **Answer:** D

*Options:* A) The Earth must lie completely within the Moon's penumbra.; B) The Moon's penumbra must touch the area where you are located.; C) The Earth must be near aphelion in its orbit of the Sun.; D) The Moon's umbra must touch the area where you are located.

*Note:* Correct option: D - The Moon's umbra must touch the area where you are located.

3. **Answer:** C

*Options:* A) 150 years; B) 200 years; C) 250 years; D) 300 years

*Note:* Correct option: C - 250 years

4. **Answer:** A

*Options:* A) size of the planet; B) presence of an atmosphere; C) distance from the sun; D) rotation period

*Note:* Correct option: A - size of the planet

5. **Answer:** D

*Options:* A) 2:3 periodic ratio of Neptune:Pluto; B) Kirkwood Gaps.; C) Gaps in Saturn's rings.; D) Breaking of small Jovian moons to form ring materials.

*Note:* Correct option: D - Breaking of small Jovian moons to form ring materials.

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'a large gap visible from Earth produced by an orbital resonance with the moon Mimas'.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'because the Moon's rotational and orbital periods are equal'.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

## Long Form Response

11. **Answer:** They actually contain a significant fraction of non-gaseous matter.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Hydrogen. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*